SACHIN KUMAR

+91 7667456574 sachinsingh.kumar07@gmail.com LinkedIn Github Portfolio Leetcode

Technical Skills

Languages & Core: Go, Python, TypeScript, JavaScript, SQL

AI & Agentic Systems: LangGraph, Multi-Agent Orchestration, Tool Calling, RAG, Agent Memory, Guardrails, Human-in-the-Loop, Context/Token Optimization

Frameworks & Runtimes: Gin, Fiber, FastAPI, NestJS, Express, Next.js,Vue.js, Angular-lonic, Node.js

Distributed Systems & Messaging: Apache Kafka, RabbitMQ, BullMQ, Apache Flink, Redis Pub/Sub, Microservices Architecture, Rate Limiting

Databases & Storage: PostgreSQL, ClickHouse, StarRocks, TimescaleDB, MongoDB, MS-SQL, Redis

DevOps, Cloud & Monitoring: Docker, AWS, GCP, CI/CD (Jenkins), Datadog, k6, SonarQube

Summary

Software Engineer with **3+ years** of **experience designing** and scaling **high-concurrency distributed systems** and **agentic AI** pipelines using **Go**, **Python**, and **Node.js.** Proven track record architecting multi-stage **LLM** agents (**LangGraph**), **low-latency** trading engines, and streaming analytics pipelines processing **millions** of events via **Kafka**, **Flink**, and **ClickHouse**. Specialized in backend performance tuning, concurrency control, and **zero-hallucination AI system design**.

Experience

Atrina Technology (Client: 1 Finance) **June 2026 - Present**

Senior Software Engineer Mumbai, Maharashtra, India

- Architected an agentic **financial** assistant in **Python** (**LangGraph**, **FastAPI**) across **7 crypto** exchanges, verified with a comprehensive suite of **73 pytest** unit and integration tests.
- Engineered a **5-stage** validation pipeline: integrated a **20B** guardrail model to filter malicious queries, a **120B** reasoning model for tool execution, and automated deterministic validators to ensure **zero hallucinated numbers**.
- Developed **10 JWT-secured Python** tools interfacing with **Go** backend services (**httpx**), enforcing strict output validation against live **DB** states.
- Reduced **LLM** token consumption by **40%** via windowed context buffering (**last 6 turns**), subsequently porting the core engine to **Node.js** with state checkpointing, per-user memory, and **human-in-the-loop** approvals.
- Maintained and optimized **7 Go** microservices using **Redis pub/sub**, eliminating goroutine leaks and controlling concurrency fan-out with worker **semaphores**.
- Developed **3 NestJS CMS** modules and supported a **570+ API** wealth-advisory platform, securing **S3** file storage using private **CloudFront** distributions and signed **URLs**.

Tech Stack: Python, LangGraph, FastAPI, Pydantic, httpx, pytest, Go, Node.js, NestJS, LangGraph Checkpointing, Retrieval-Augmented Generation (RAG), AI Agents, LLM Tool Calling, Guardrails, Agent Memory, Human-in-the-Loop (HITL), JWT Authentication, Redis Pub/Sub, AWS S3, Amazon CloudFront.

Winsoft Technologies (Client: Choice Broking) **Oct 2025 - June 2026**

Software Developer Mumbai, Maharashtra, India

- Built core **Go** (**Gin, GORM**) backend services for an unlisted equity trading platform, implementing **ACID-compliant** order matching, placement, and settlement on **MS-SQL**.
- Engineered an asynchronous **notification** service processing **millions** of messages via **RabbitMQ** and **Kafka**, integrating **Apache Flink** to stream real-time trade data into **StarRocks** and **ClickHouse** for sub-second analytics.
- Decreased **API** latency by **30%** through targeted **Redis** caching and fine-grained concurrency control (**goroutines, channels, sync.Mutex**), actively profiling **p95/p99** latencies using **Datadog** and **k6** load tests.
- Implemented an in-memory token-bucket **rate limiter** in **Go**, providing burst handling without incurring **Redis** network round-trip overhead.
- Developed a high-throughput courier **tracking system** in **Go** and **Next.js**, processing **5,000–10,000** real-time status dispatches daily via **RabbitMQ**.
- Built automated internal **microservices** in **FastAPI**, including an automated invoice generator producing **~1,000** bills daily and a custom pull-request security scanner integrated with **SonarQube**.

Tech Stack: Go, Gin, GORM, Python, FastAPI, Next.js, Microsoft SQL Server (MS SQL), Redis, Apache Kafka, RabbitMQ, Apache Flink, StarRocks, ClickHouse, Goroutines, Channels, Mutexes, Semaphores, Token Bucket Rate Limiting, Datadog, k6, SonarQube.

Pivotroots **Aug 2023 - Oct 2025**

Software Developer Mumbai, Maharashtra, India

- Built an **offline-first** mobile-friendly web application using **SQLite** and an **asynchronous** sync queue, boosting field-agent operational efficiency by **40%** backed by **20+ Go/Node.js REST** endpoints.
- **Automated** scheduled batch **notifications** and recurring jobs, reducing operational workload by **30%**, and created a role-based image validation pipeline reducing lead review cycles by **20%**.
- Deployed an **AI-powered OCR** and vision verification system for vehicle odometer and breathalyzer telemetry, decreasing manual data entry errors by **10%**.
- Decoupled heavy background compute workloads onto **RabbitMQ** queues to lower primary database contention by **10%**, utilizing **GCP** Cloud Scheduler and **Google Maps APIs** for **live fleet tracking**.

Tech Stack: Go, Node.js, REST APIs, SQLite, RabbitMQ, Google Cloud Platform (GCP), Google Maps APIs, Google Cloud Scheduler, Offline-First Architecture, Asynchronous Job Processing, Task Scheduling, Real-Time Driver Tracking.

Education & Certifications

- **Veer Kunwar Singh University —** B.Sc. (Hons) in Mathematics | Sep 2018 – Sep 2022
- **Masai School —** Certification in Full Stack Web Development | Feb 2022 – Feb 2023